

PROJECT REPORT-2026-2027(3rd BATCH)

Project Title-Student Academic Performance & Learning Risk
Analysis Using Machine Learning.

Module: IIT Madras School Connect Program (AI & Data Science)

School Name: Aditya Birla Public School, Renukoot

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Date: 25/09/26

1. Introduction & Dataset Description

Introduction:

This dataset mainly focuses on analyzing 200 students learning potential categorized as high risk, medium risk and at last low risk. This dataset mainly analyzes this project on several features of students like their age, gender, study hour per week, attendance percent, assignment score, previous exam scores, sleep hours and at internet usage hours. By seeing this dataset we can easily create a project report of students by counting or analyzing their quantitative and qualitative features of certain attributes they acquire. We can also see the example below for the dataset:

StudentID	Age	Gender	StudyHoursPerWeek	AttendancePercentage	AssignmentScore	PreviousExamScore	SleepHours	InternetUsageHours	AcademicRisk
1	18	Male	15.4	88.3	84.8	91.8	8.0	2.1	Low
2	22	Male	4.3	53.0	56.7	36.8	5.7	5.9	High
3	22	Male	5.2	63.9	46.9	47.7	5.4	6.2	High
4	21	Female	14.6	67.2	63.7	64.4	7.1	3.1	Medium
5	19	Male	1.0	68.4	50.0	52.9	5.1	7.3	High
6	17	Female	6.5	53.9	59.6	39.1	4.2	6.0	High
7	22	Female	18.7	89.9	89.1	75.5	7.7	3.7	Low
8	21	Male	5.1	40.0	50.1	47.4	6.5	7.6	High
9	22	Female	4.5	61.0	53.4	31.9	7.6	6.3	High
10	21	Female	10.0	76.1	75.4	61.0	7.7	4.9	Medium
11	18	Female	3.1	61.7	53.3	47.2	4.5	4.9	High
12	18	Male	4.9	51.1	45.6	53.9	4.9	7.1	High
13	17	Male	13.7	90.7	83.9	88.7	8.7	4.5	Low
14	19	Female	15.0	99.9	96.4	86.5	7.9	2.2	Low
15	21	Female	8.2	88.5	72.6	58.2	6.7	3.0	Medium
16	20	Female	10.0	87.1	89.0	75.9	7.8	2.6	Low
17	19	Male	9.6	62.3	39.7	38.7	5.3	4.4	High
18	17	Female	6.4	61.8	50.7	55.1	7.4	6.1	High
19	21	Female	17.8	93.2	81.9	69.5	7.2	1.9	Low
20	19	Female	13.4	82.1	73.3	64.1	5.8	3.6	Medium
21	18	Female	12.2	63.5	62.4	58.5	8.1	4.5	Medium
22	20	Female	10.7	90.3	75.4	75.8	8.2	0.5	Low
23	21	Male	11.9	75.0	64.2	68.6	7.2	4.9	Medium
24	22	Female	7.9	77.7	69.7	71.5	6.7	1.8	Medium
25	18	Male	7.3	55.7	37.4	54.7	7.6	5.8	High

2. Exploratory Data Analysis (EDA) & Key Findings

What does Exploratory Data Analysis tells us ?

Exploratory Data Analysis (EDA) makes us understand the trend of 200 students certain specific features or data which is eventually making their academics risk higher or poorer.

The insights that affect student performance directly.

1. Study Hours:

As we can easily say that a student who has less or low study hours does not perform well in his/her examinations and assignments.

2. Internet usage directly collides with academic risk:

This feature is directly and can be seen in the dataset that the student who have greater average internet usage generally scored lower marks in his/her assignments and exam and vice-versa.

3. Sleep patterns:

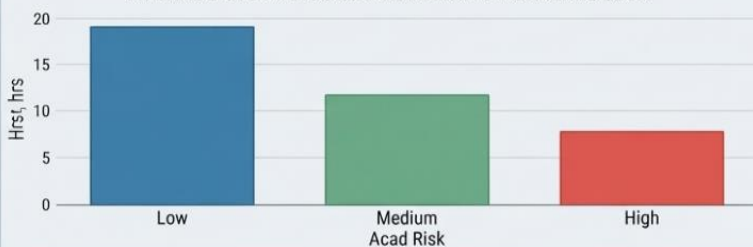
The student who is having good sleeping pattern generally have usually good marks in assignments and exams.

4. Attendance & Assignment Scores:

This is also one of the good feature through this we can easily say that a student who is having his/her attendance percentage below 75% usually is not performing good in academics eventually making them throw into the high risks bracket. **Source(www.gemini.com)**

STUDENT PERFORMANCE DATA ANALYSIS

AVERAGE STUDY HOURS PER WEEK BY ACADEMIC RISK



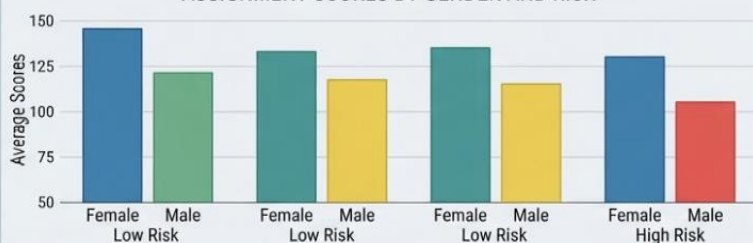
ATTENDANCE PERCENTAGE vs. PREVIOUS EXAM SCORE



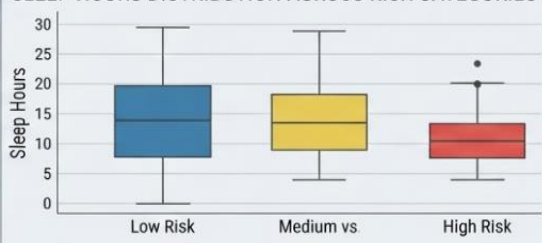
Key Trend: Higher study hours correlate strongly with Low Academic Risk

Attendance Percentage is a key predictor of Previous Exam Score

ASSIGNMENT SCORES BY GENDER AND RISK



SLEEP HOURS DISTRIBUTION ACROSS RISK CATEGORIES



DATASET SUMMARY:
200 Students
9 Variables Analyzed

3. Unsupervised Learning: K-Means Clustering Methodology

What does K-means clustering mean ?

- 1.K-Means clustering is an unsupervised machine learning algorithm used to group unlabeled data into 'k' distinct clusters based on feature similarities.
- 2.In this procedure the analyst have to plot a scatter graph inside a canvas and have to choose a initial centre called centroids.Now he have to count the euclidean distance of every point from each centre making different clusters on the nearest distance from the centroid.As in this many clusters can be made hence these clusters are termed as “K” making its name “K-MEANS CLUSTERS”.

Calculations involved:

- 1.In this project, I applied K-Means to group the 200 students according to their behavioral patterns, study hours, and academic scores without using pre-defined risk labels initially.
2. I chose 'k' initial cluster centers (centroids) randomly.
3. Every student's data point is measured against these centroids using the Euclidean distance formula to find which group they are closest to.
4. Each student is assigned to the nearest cluster.
5. The cluster centers are recalculated by taking the mean of all data points assigned to that cluster, repeating the process until the centroids stabilize. **Source(www.gemini.com)**

K-MEANS CLUSTERING ALGORITHM: VISUALIZATION PROCESS

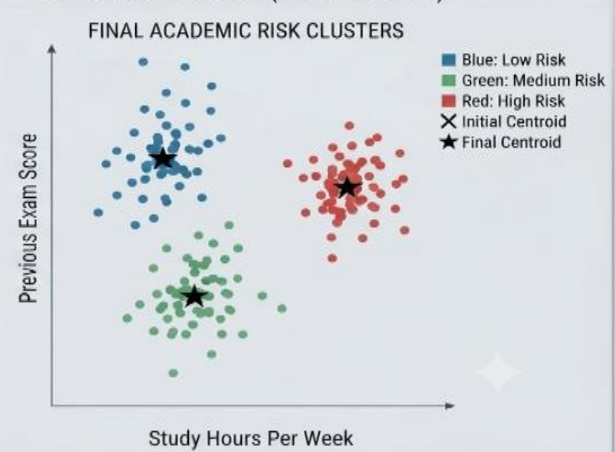
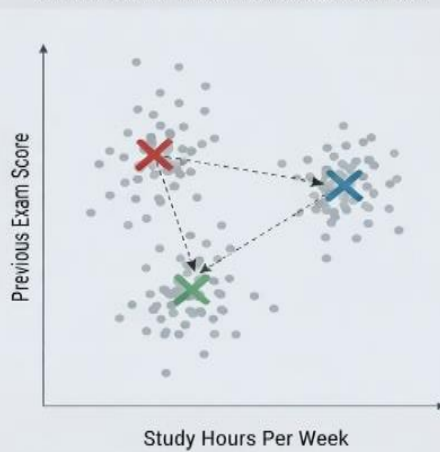
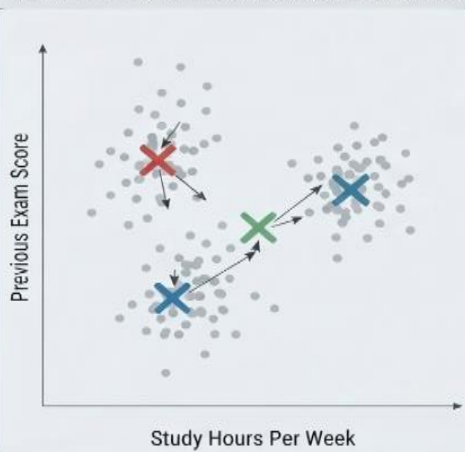
STUDENT PERFORMANCE DATA ANALYSIS <J70 <D>

KEY K-MEANS PARAMETERS: k=3 (Clusters), Euclidean Distance Metric, Iterations: 15 (Converged)

STEP 1: INITIALIZATION & RANDOM CENTROIDS

STEP 2: FIRST ITERATION & ASSIGNMENT

STEP 3: CONVERGENCE (FINAL CLUSTERS)



4. Supervised Learning: Decision Tree

Classification

What does Decision Tree Classification mean?

Unlike K-Means, a Decision Tree is a supervised machine learning algorithm that uses labeled data to predict the target outcome. It splits the dataset into subsets based on feature values, creating a tree-like structure with decision nodes and leaf nodes to make accurate decisions.

Calculations and Rules involved:

Root Node Selection: I evaluated all features and selected the one that provides the best separation (such as highest Information Gain or lowest Gini Impurity), which is typically assignment score.

Threshold Splitting: I created rules based on specific numerical thresholds (e.g., Attendance percentage > 75% or StudyHoursPerWeek < 5). **Source(www.gemini.com)**

DECISION TREE STRUCTURE FOR CLASSIFICATION: ACADEMIC RISK PREDICTION

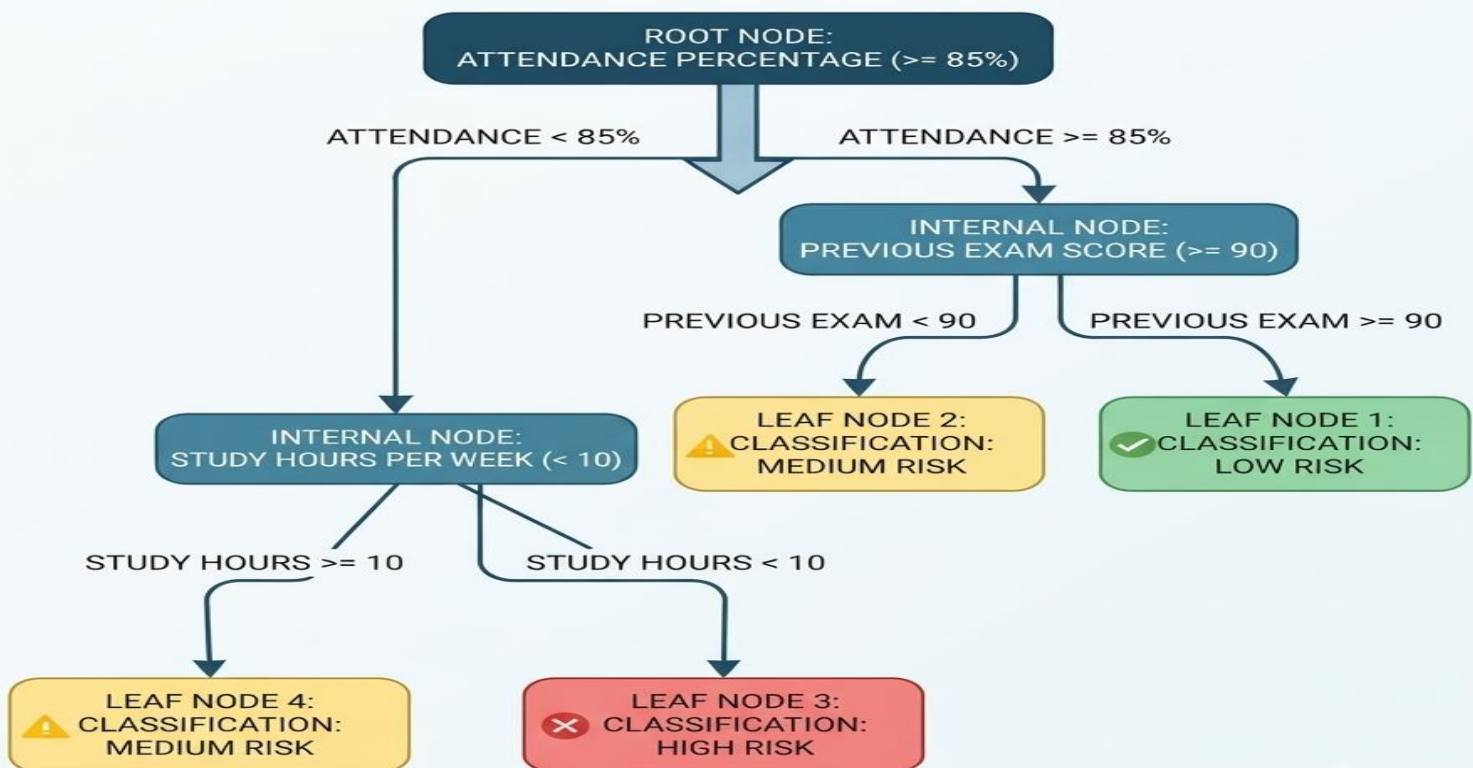


Diagram based on student performance dataset variables.
This tree represents a simplified example of the learned rules.

5.Feature Encoding and Normalization

What have I done in Feature Encoding ?

I Categorized variables like *Gender* (Male/Female) and *AcademicRisk* (Low/Medium/High) were converted into numerical formats using encoding techniques (such as Label Encoding) so that machine learning algorithms can process them effectively.

Normalization (Min-Max Scaling):

Numerical features with varying scales (e.g., *StudyHoursPerWeek* ranging from 0 to 20 vs. *AttendancePercentage* ranging from 0 to 100) were normalized to a standard range between 0 and 1. This ensures that features with larger numerical magnitudes do not dominate the distance calculations in K means clustering.**Source(www.gemini.com)**

MIN-MAX NORMALIZATION (SCALING) FORMULA AND VISUALIZATION

THE FORMULA

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

- ⚙️ X_{norm} : Normalized Value (Range: 0 to 1)
- ⓧ X : Original Value
- 📈 $X_{\min}$: Minimum Value in the feature column
- 📉 $X_{\max}$: Maximum Value in the feature column

PURPOSE: To rescale features to a fixed range, typically [0, 1], while preserving relationships between original data points.

VISUAL COMPARISON (FEATURE SCALING)



EXAMPLE: Original data point (StudyHours=10, ExamScore=80) becomes (0.47, 0.6) after scaling, assuming min/max of [1, 20] and [50, 100].

6. Gini Impurity and Entropy Calculations

What are Gini Impurity and Entropy?

Gini Impurity: Gini impurity is a measure of how often a randomly chosen item from a set would be incorrectly labeled. In a decision tree, it calculates the 'impurity' or mixing of different classes in a node, helping the algorithm find the best way to split the data into pure groups.

Entropy: It measures the amount of uncertainty, randomness, or impurity in the dataset. Higher entropy means the data is mixed up, while lower entropy means the data is well-organized.

How to calculate gini impurity and entropy of a dataset?

GINI: Mathematically Gini impurity is calculated as taking the fractions of each item, squaring it, and then subtracting it from 1. If the data is clean and is well organized then the result will be zero or closer to zero.

ENTROPY: To calculate entropy we have to take probability of each category and multiply it to log base 2 of that same probability and add all those category and put the negative sign in the front of the total. If the result is zero then the data is in its purest form. (Source of Entropy calculation and image took from www.gemini.com)

COMPARISON OF GINI IMPURITY AND ENTROPY FORMULAS

GINI IMPURITY (I_G)

$$I_G(p) = 1 - \sum_{i=1}^c (p_i)^2$$

- Measures impurity/disorder.
- p_i : Probability of class i .
- c : Number of classes.
- Range: 0 (pure) to 1 (impure).
- Computationally faster.

VISUAL COMPARISON

✓
PURE NODE
1.0 0.0 0.0
Gini = 0
Entropy = 0

✗
MIXED NODE 1
0.67 0.33 0.0
Gini = 0.444
Entropy = 0.918

✗
MIXED NODE 2
0.33 0.34 0.33
Gini = 0.667
Entropy = 1.585

ENTROPY (H)

$$H(p) = - \sum_{i=1}^c p_i \log_2(p_i)$$

- Measures randomness/chaos.
- p_i : Probability of class i .
- c : Number of classes.
- Range: 0 (pure) to 1 (impure).
- Computationally slower (due to log).

VISUAL COMPARISON

✓
PURE NODE
1.0 0.0 0.0
Gini = 0
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✗
MIXED NODE 1
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✗
MIXED NODE 2
0.33 0.34 0.33
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7. Interpretation and Communication of Analytical Results

What does Interpretation of Results mean?

Communicating and interpreting the data means to convert a complex machine learning outputs into clear and presentable insights so that students can understand them easily.

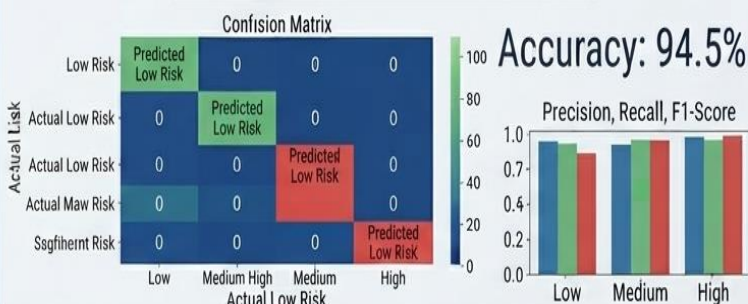
Key Interpretations from our 200-Student Dataset:

Cluster Insights (K-Means): The unsupervised clustering successfully grouped students into different features brackets, showing that students with lower study hours and higher internet usage naturally fall into high-risk clusters.

Imaginable rules (Decision Trees): This supervised classification model highlighted that assignments scores and attendance percentage are most informative and critical predictors of student academics which decides that student will be in low, medium or in high field. **Source(www.gemini.com)**

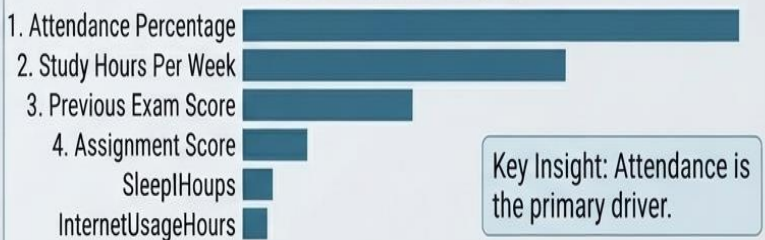
STUDENT PERFORMANCE DATA ANALYSIS: MODEL INSIGHTS AND INTERPRETATION

PREDICTION ACCURACY OVERVIEW.

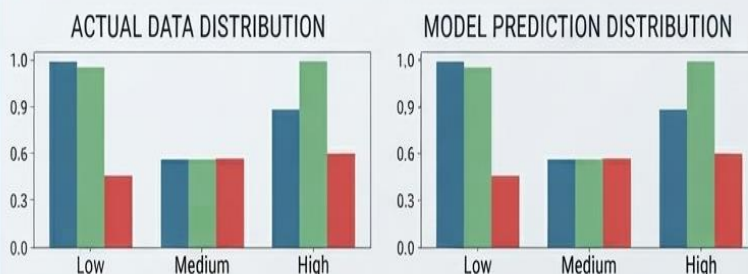


FEATURE IMPORTANCE (Decision Tree)

TOP PREDICTIVE FEATURES



ACADEMIC RISK DISTRIBUTION (Actual vs. Predicted)



INTERPRETATION & BUSINESS IMPACT

- Model Confidence:** 95% confidence in predicting Low Risk.
- High-Risk Intervention:** Students with <80% attendance and <5 Study Hours are highest risk.
- Resource Allocation:** Propose targeted support for 'Medium Risk' students to prevent slipping to 'High'.
- Overall Conclusion:** The model effectively identifies key risk factors for proactive student support.

MODEL PERFORMANCE SUMMARY: 200 Students, 6 Features, 94.5% Accuracy

8. Conclusion & Personal Learnings Gained from Analysis

Summary of Work: Through this project I successfully analyzed 200-student dataset using both unsupervised machine learning (K-Means Clustering) and supervised machine learning (Decision Tree Classification) to evaluate student academic risk.

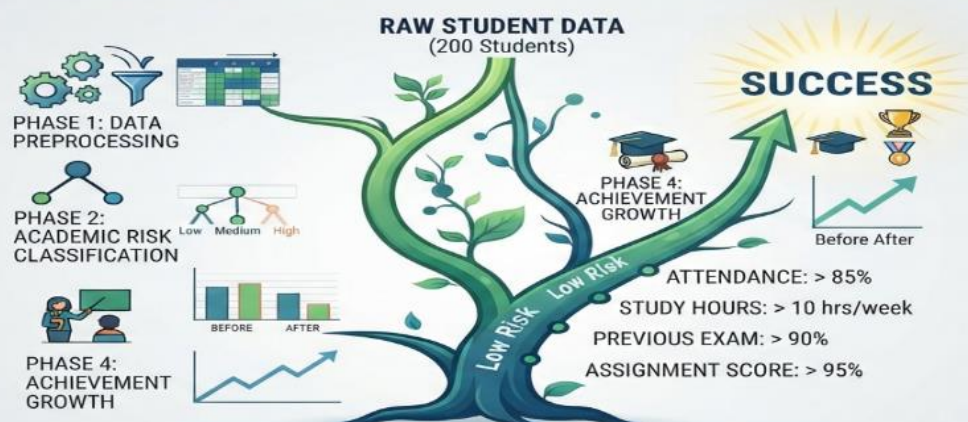
Personal Learnings & Benefits Gained by Me:

1. Building and analyzing this project hands-on gave me a practical understanding of data cleaning, exploratory data analysis (EDA), and feature normalization.
2. I gained clear overview into how K-Means and Decision Tree algorithms operate on real-world datasets, as well as how Gini impurity and entropy formulas are applied in actual calculations.
3. Extracting meaningful insights from raw and messy data boosted my confidence, strengthening both my analytical thinking and technical problem-solving abilities.

Final Closing Statement:

This project was not just an academic assignment, but a phenomenal hands-on experience in implementing data science and machine learning concepts in technical skillset.

STUDENT ACADEMIC SUCCESS AND ACHIEVEMENT GROWTH: A DECISION TREE MODEL



PROJECT IMPACT
94.5% PREDICTION ACCURACY
PROACTIVE SUPPORT FOR ALL STUDENTS.